Signals and Systems
Assignment 1: solution

Chadi Abou-Rjeily

Department of Electrical and Computer Engineering
Lebanese American University
chadi.abourjeily@lau.edu.lb

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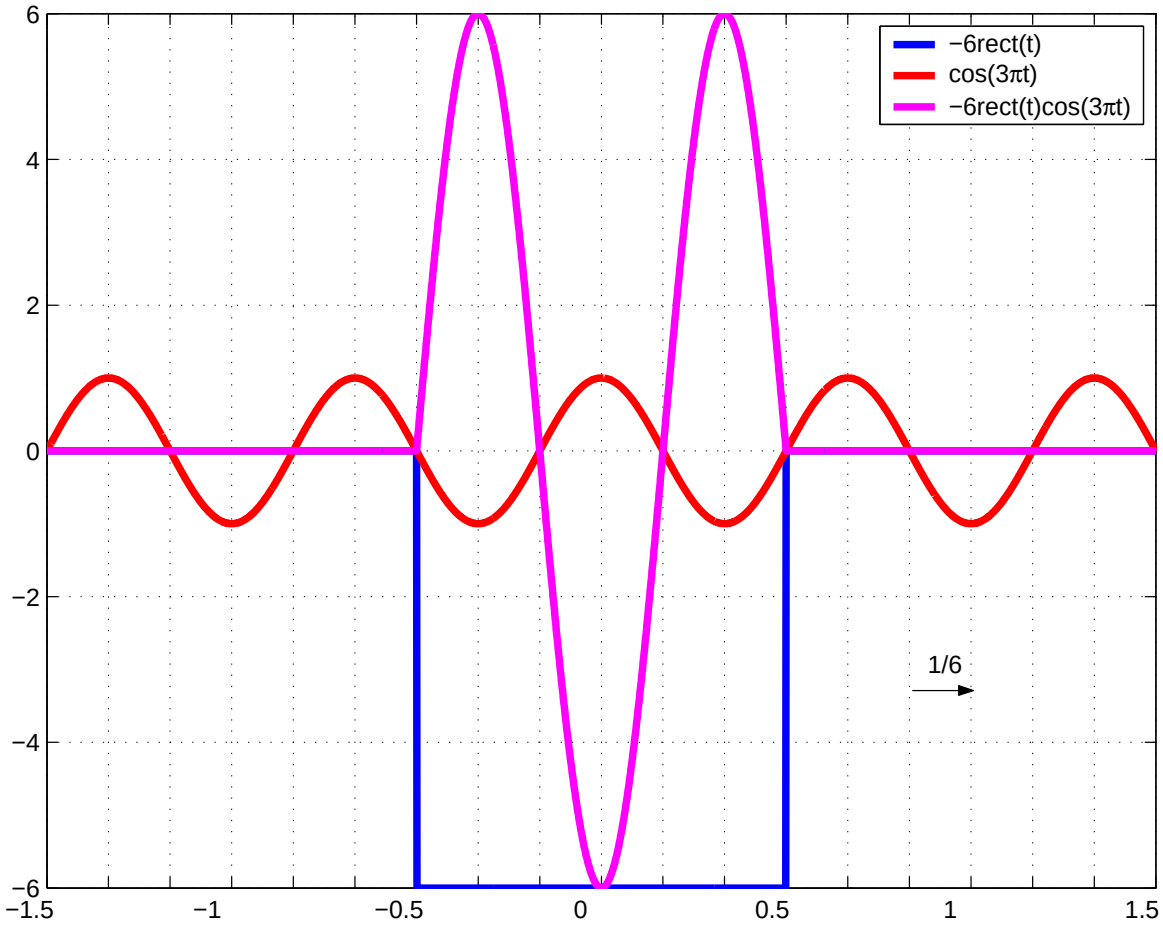
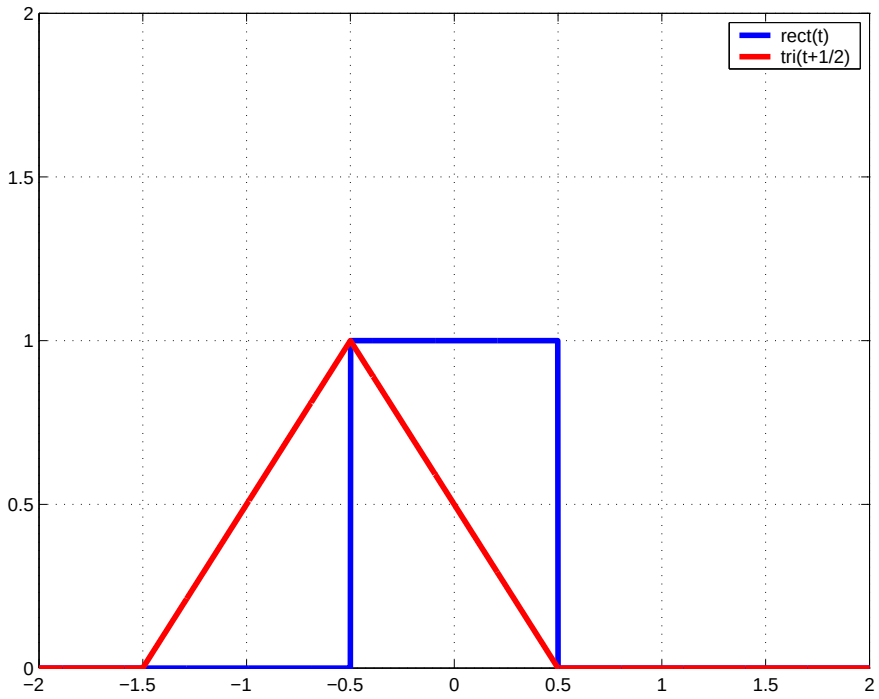
Exercise 1 (1)

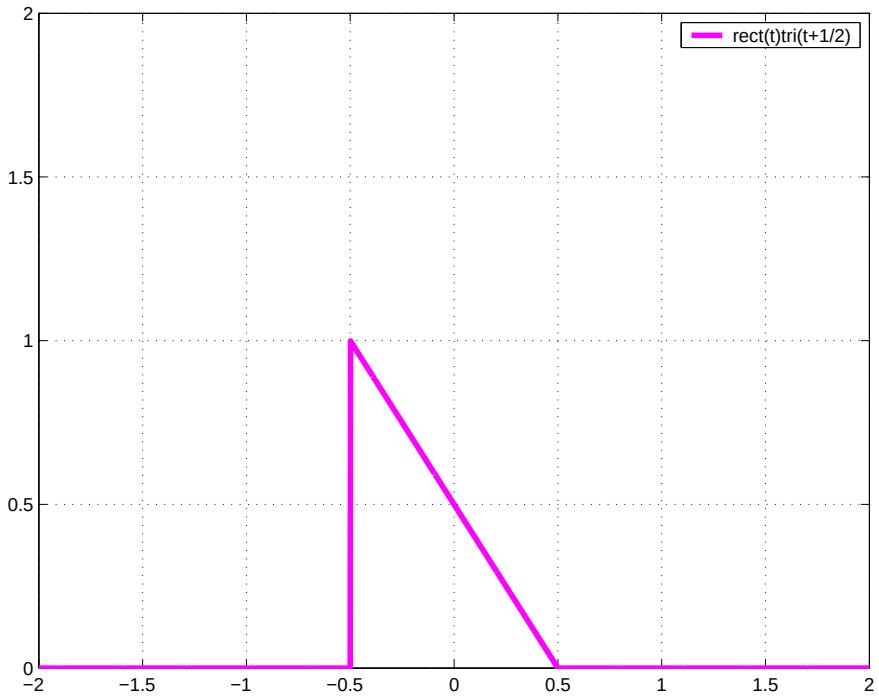
Plot the following signal: $f(t) = -6\mathrm{rect}(t)\cos(3\pi t)$



Exercise 1 (2)

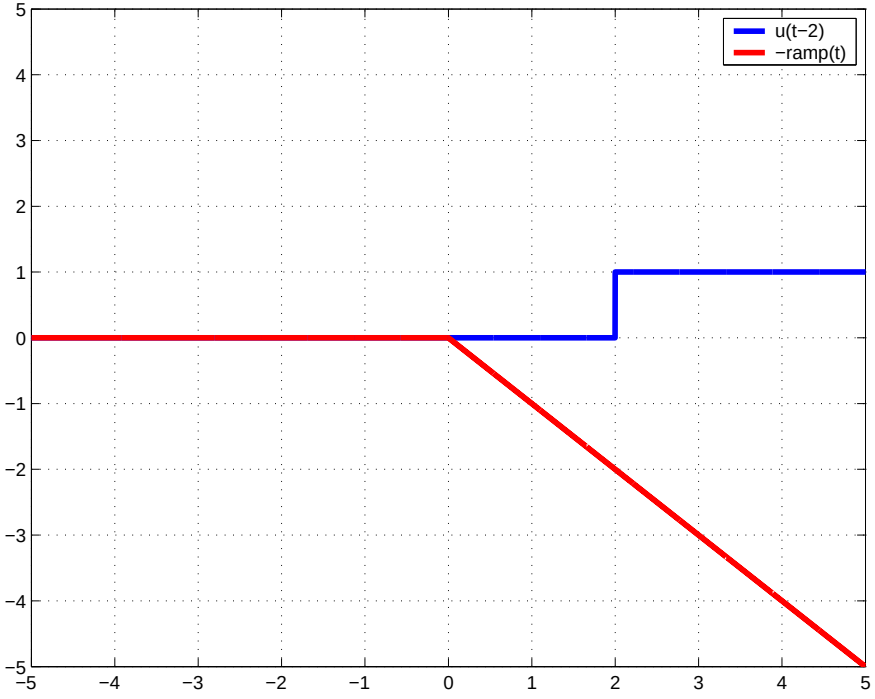
Plot the following signal: $f(t) = \mathrm{rect}(t)\mathrm{tri}(t + \frac{1}{2})$

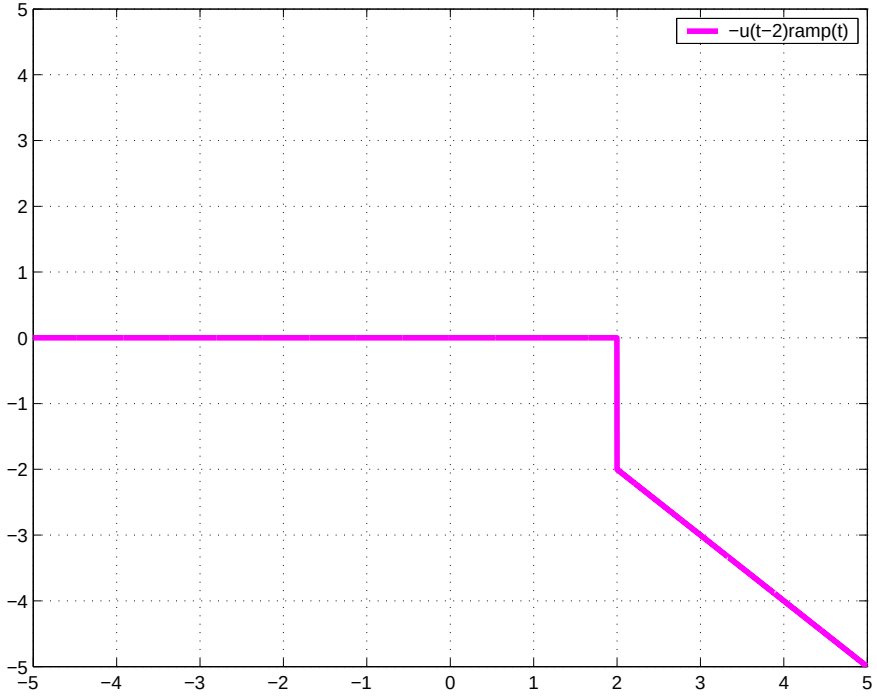




Exercise 1 (3)

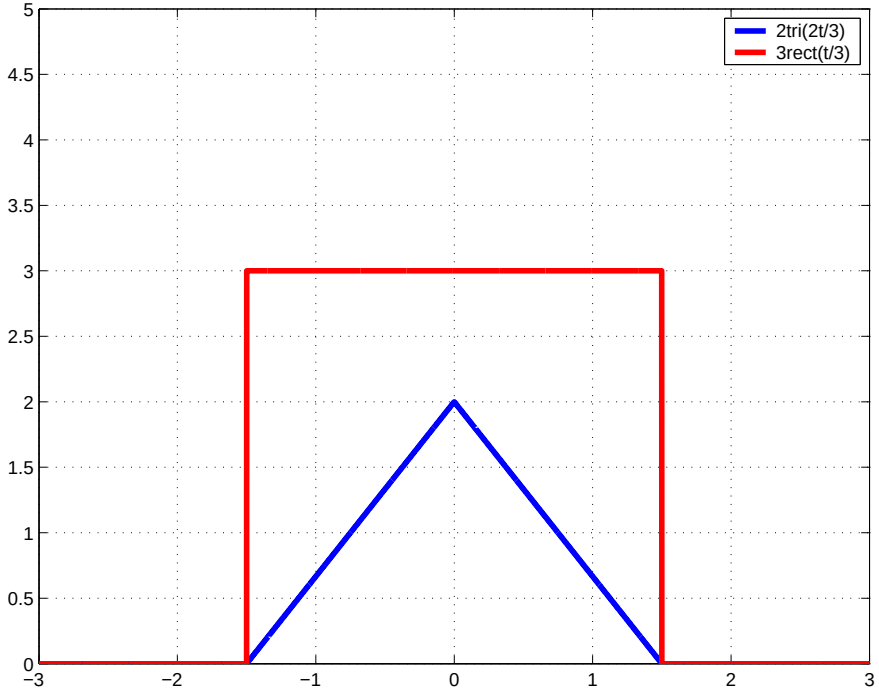
Plot the following signal: $f(t) = -\mathrm{ramp}(t)\mathrm{u}(t-2)$

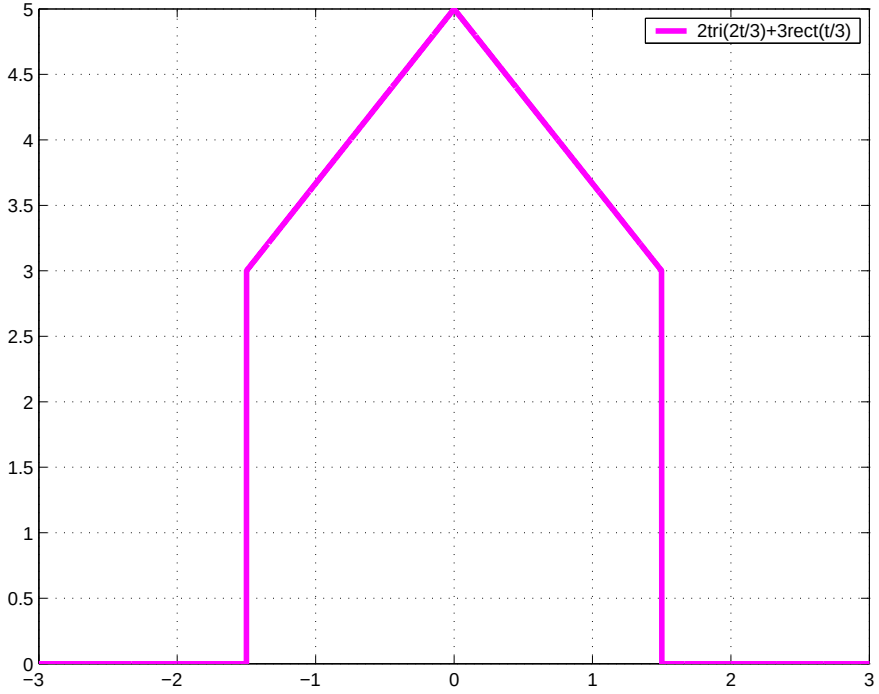




Exercise 1 (4)

Plot the following signal: $f(t) = 2\mathrm{tri}(\frac{2t}{3}) + 3\mathrm{rect}(\frac{t}{3})$





Exercise 2 (1)

Linearity of $y(t) = x(t-1) - x(t-5)$:

$$\begin{aligned} x_1(t) &\rightarrow y_1(t) = x_1(t-1) - x_1(t-5) \\ x_2(t) &\rightarrow y_2(t) = x_2(t-1) - x_2(t-5) \end{aligned}$$

- For $x(t) = \alpha x_1(t) + \beta x_2(t)$:

$$\begin{aligned} y(t) &= x(t-1) - x(t-5) \\ &= [\alpha x_1(t-1) + \beta x_2(t-1)] - [\alpha x_1(t-5) + \beta x_2(t-5)] \\ &= \alpha \underbrace{[x_1(t-1) - x_1(t-5)]}_{=y_1(t)} + \beta \underbrace{[x_2(t-1) - x_2(t-5)]}_{=y_2(t)} \\ &= \alpha y_1(t) + \beta y_2(t) \end{aligned}$$

- Consequently, this system is linear.

Exercise 2 (2)

Linearity of $y(t) = |t|x(t)$:

$$\begin{aligned} x_1(t) &\rightarrow y_1(t) = |t|x_1(t) \\ x_2(t) &\rightarrow y_2(t) = |t|x_2(t) \end{aligned}$$

- For $x(t) = \alpha x_1(t) + \beta x_2(t)$:

$$\begin{aligned} y(t) &= |t|x(t) \\ &= |t|\left[\alpha x_1(t) + \beta x_2(t)\right] \\ &= \alpha \underbrace{|t|x_1(t)}_{y_1(t)} + \beta \underbrace{|t|x_2(t)}_{y_2(t)} \\ &= \alpha y_1(t) + \beta y_2(t) \end{aligned}$$

- Consequently, this system is linear.

Exercise 2 (3)

Linearity of $y(t) = (x(t))^2 - x(t-1)$:

$$\begin{aligned} x_1(t) &\rightarrow y_1(t) = (x_1(t))^2 - x_1(t-1) \\ x_2(t) &\rightarrow y_2(t) = (x_2(t))^2 - x_2(t-1) \end{aligned}$$

- For $x(t) = \alpha x_1(t) + \beta x_2(t)$:

$$\begin{aligned} y(t) &= (x(t))^2 - x(t-1) \\ &= [\alpha x_1(t) + \beta x_2(t)]^2 - [\alpha x_1(t-1) + \beta x_2(t-1)] \\ &= \alpha^2 (x_1(t))^2 - \alpha x_1(t-1) + 2\alpha\beta x_1(t) x_2(t) + \beta^2 (x_2(t))^2 - \beta x_2(t-1) \end{aligned}$$

- On the other hand:

$$\alpha y_1(t) + \beta y_2(t) = \alpha (x_1(t))^2 - \alpha x_1(t-1) + \beta (x_2(t))^2 - \beta x_2(t-1)$$

- Consequently, $y(t) \neq \alpha y_1(t) + \beta y_2(t)$ and the system is not linear.

Exercise 2 (4)

Time-invariance of $y(t) = x(t-1) - x(t-5)$:

- $y(t - t_0)$ is given by:

$$y(t - t_0) = x(t - t_0 - 1) - x(t - t_0 - 5)$$

- On the other hand, when the input is equal to $X(t) = x(t - t_0)$ then:

$$\begin{aligned} Y(t) &= X(t-1) - X(t-5) \\ &= x(t - t_0 - 1) - x(t - t_0 - 5) \\ &= y(t - t_0) \end{aligned}$$

- Consequently, this system is time-invariant.

Exercise 2 (5)

Time-invariance of $y(t) = |t|x(t)$:

- $y(t - t_0)$ is given by:

$$y(t - t_0) = |t - t_0|x(t - t_0)$$

- On the other hand, when the input is equal to $X(t) = x(t - t_0)$ then:

$$\begin{aligned} Y(t) &= |t|X(t) \\ &= |t|x(t - t_0) \\ &\neq y(t - t_0) \end{aligned}$$

- Consequently, this system is not time-invariant.

Exercise 2 (6)

Time-invariance of $y(t) = (x(t))^2 - x(t-1)$:

- $y(t - t_0)$ is given by:

$$y(t - t_0) = (x(t - t_0))^2 - x(t - t_0 - 1)$$

- On the other hand, when the input is equal to $X(t) = x(t - t_0)$ then:

$$\begin{aligned} Y(t) &= (X(t))^2 - X(t-1) \\ &= (x(t - t_0))^2 - x(t - t_0 - 1) \\ &= y(t - t_0) \end{aligned}$$

- Consequently, this system is time-invariant.

Exercise 2 (7)

Causality of $y(t) = x(t-1) - x(t-5)$:

- The output at time t depends on the inputs at times $t-1$ and $t-5$. That is, it does not depend on future inputs.
- Consequently, this system is causal.

Causality of $y(t) = |t|x(t)$:

- The output depends only on the present input.
- Consequently, this system is causal.

Causality of $y(t) = (x(t))^2 - x(t-1)$:

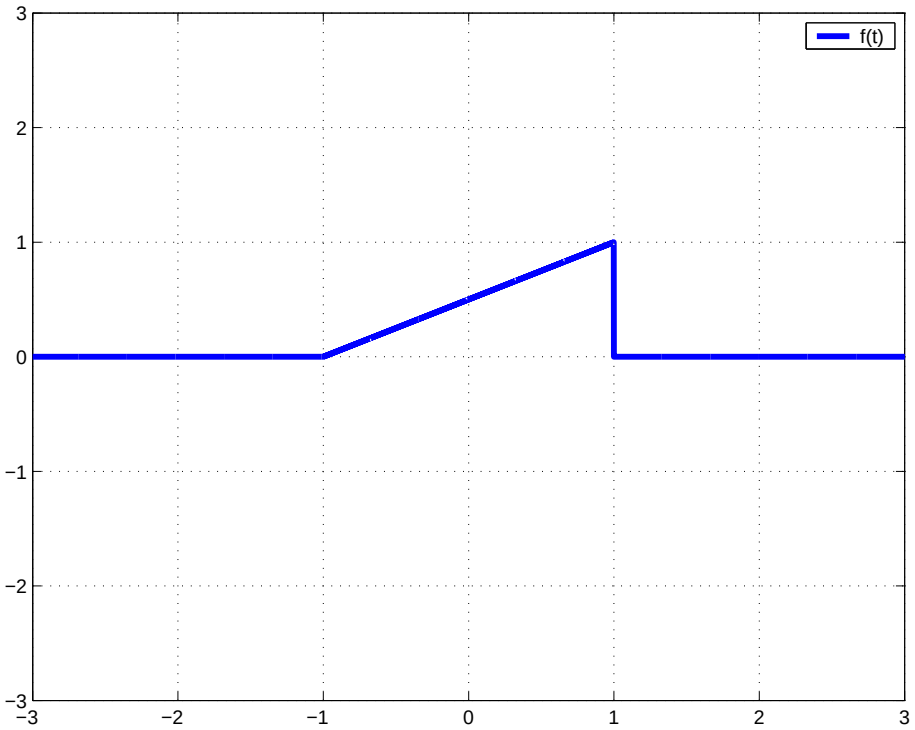
- The output at time t depends on the inputs at times t and $t-1$.
- Consequently, this system is causal.

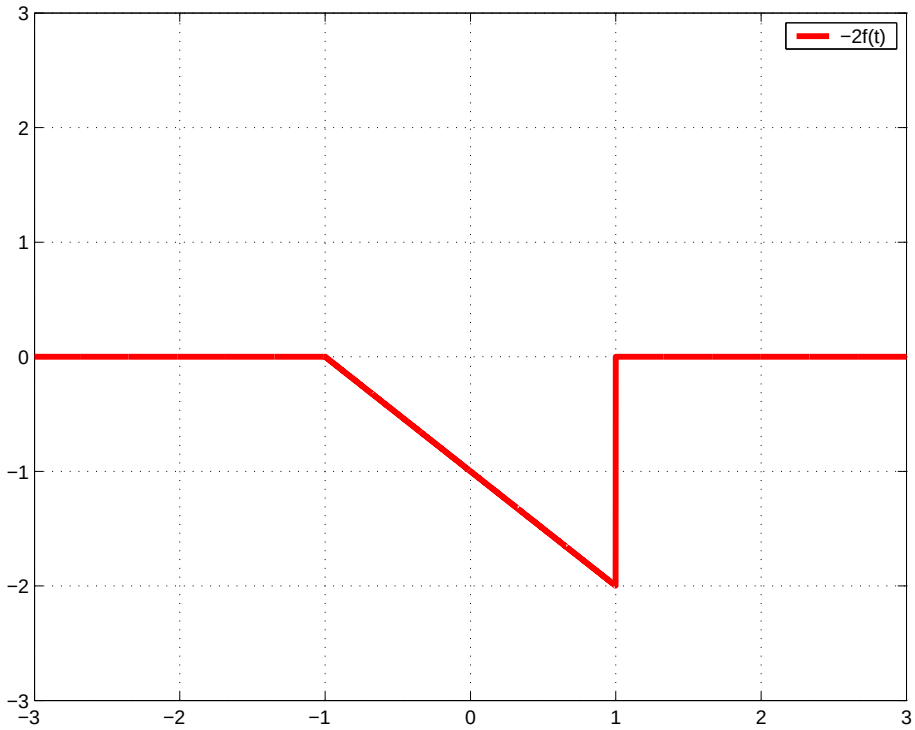
For example:

- $y(t) = 2x(t) + x(t+1)$ is not causal.
- $y(t) = x(1-t)$ is not causal. In fact, the output will depend on future inputs when $t < 1-t$ implying that $t < \frac{1}{2}$. For example, the output at time $t = 0$ depends on the input at time $1-t = 1$ and the system is not causal.

Exercise 3 (1)

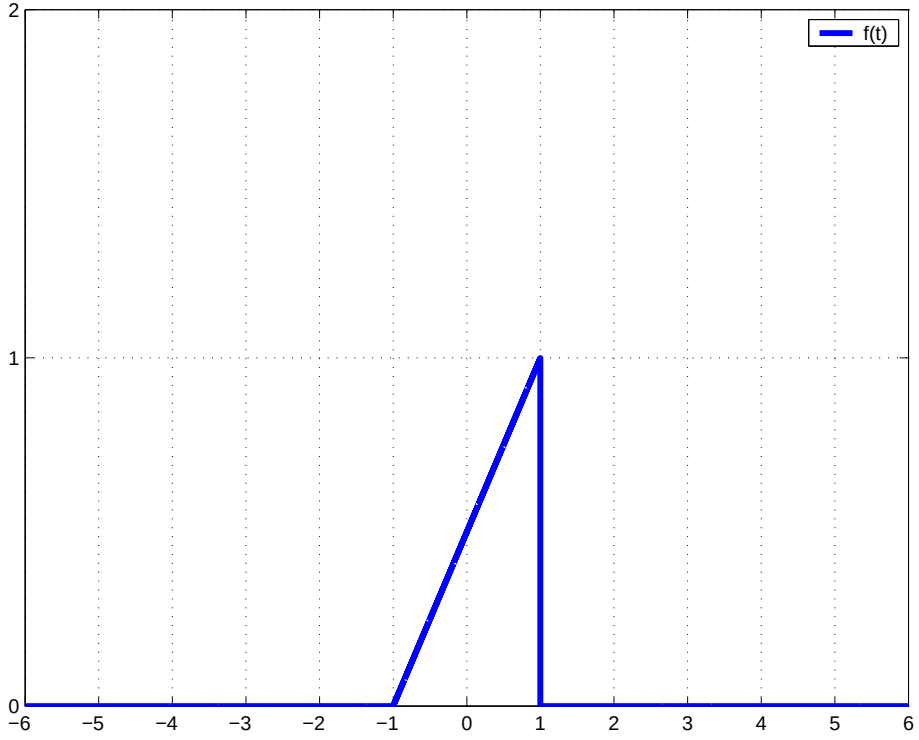
Plot: $-2f(t)$

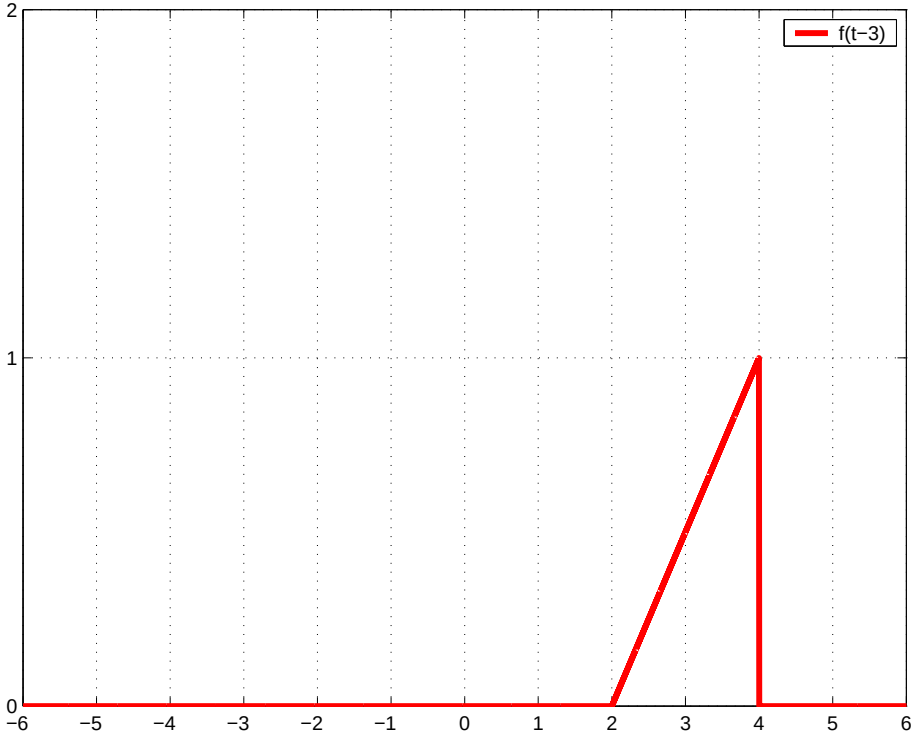




Exercise 3 (2)

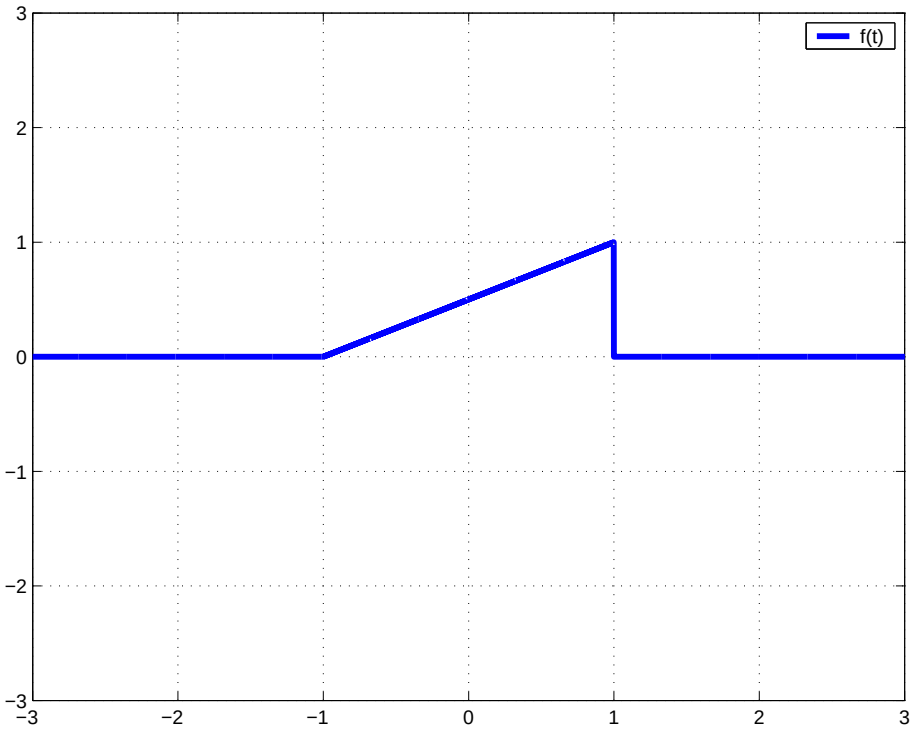
Plot: $f(t-3)$

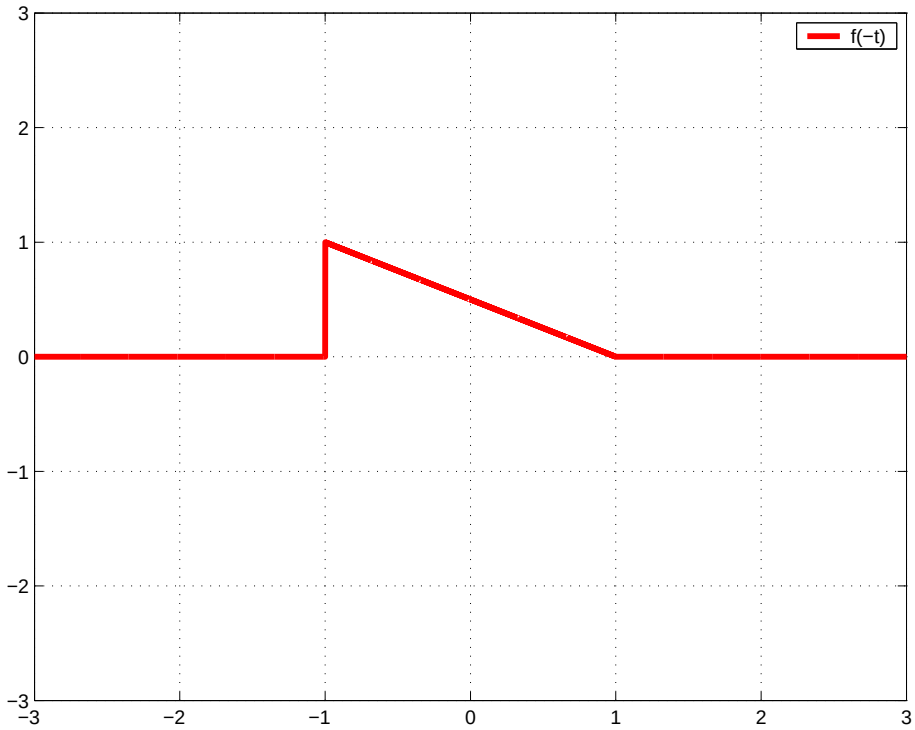




Exercise 3 (3)

Plot: $f(-t)$

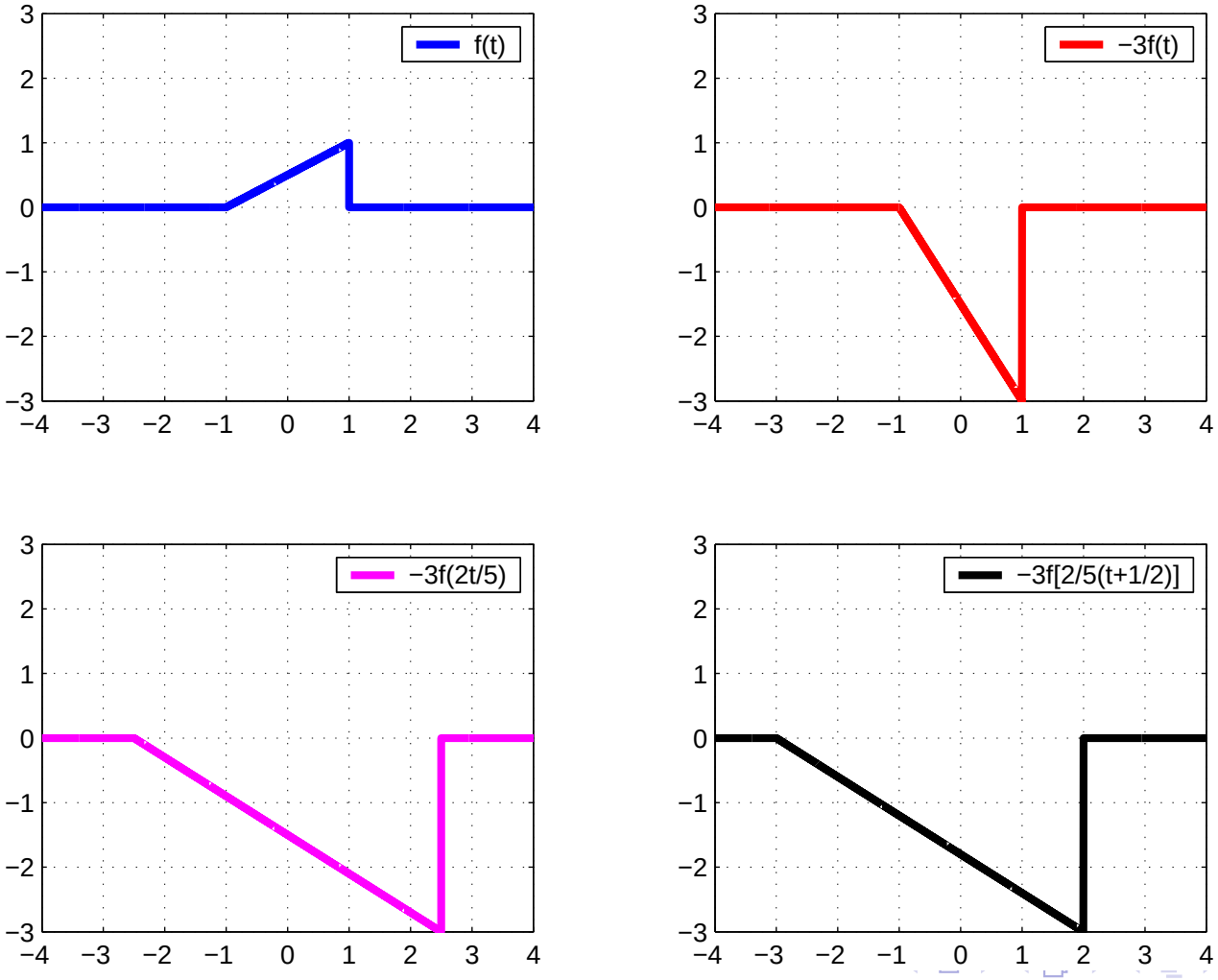




Exercise 3 (4)

Plot: $-3f(\frac{2t+1}{5})$

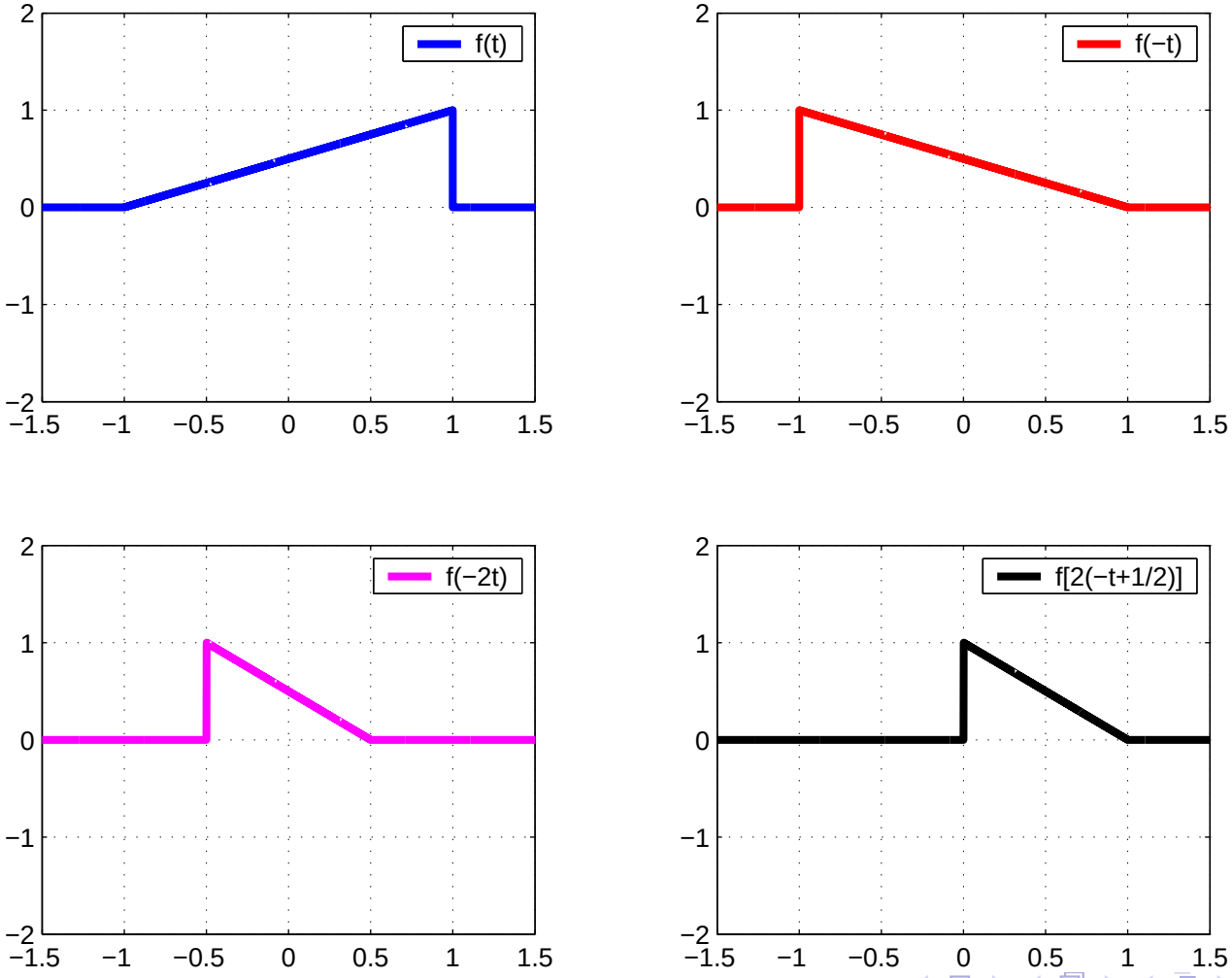
$$-3f\left(\frac{2t+1}{5}\right) = -3f\left(\frac{2}{5}(t+\frac{1}{2})\right)$$



Exercise 3 (5)

Plot: $f(-2t+1)$

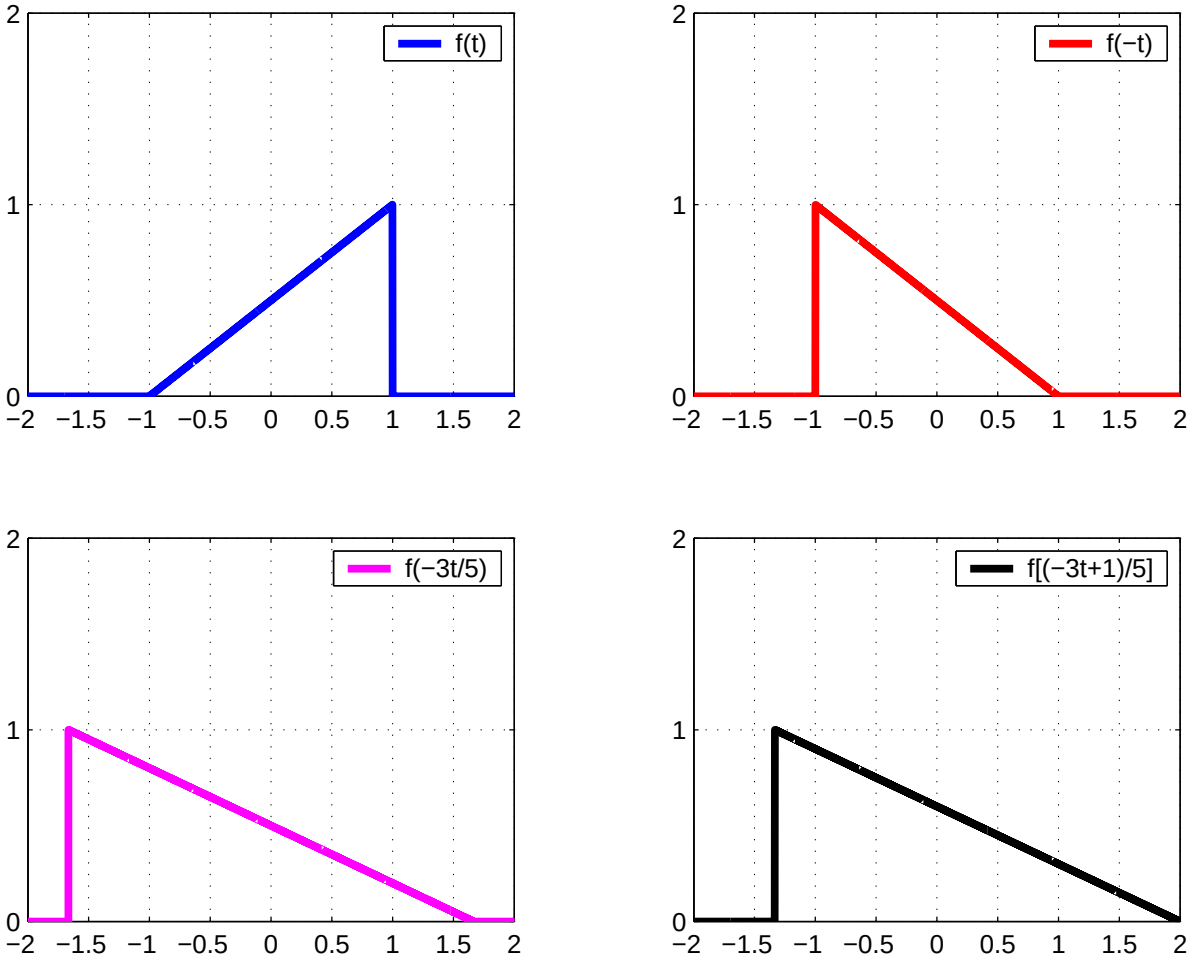
$$f(-2t+1) = f\left(2(-t+\frac{1}{2})\right)$$



Exercise 3 (6)

Plot: $f\left(\frac{-3t+1}{5}\right)$

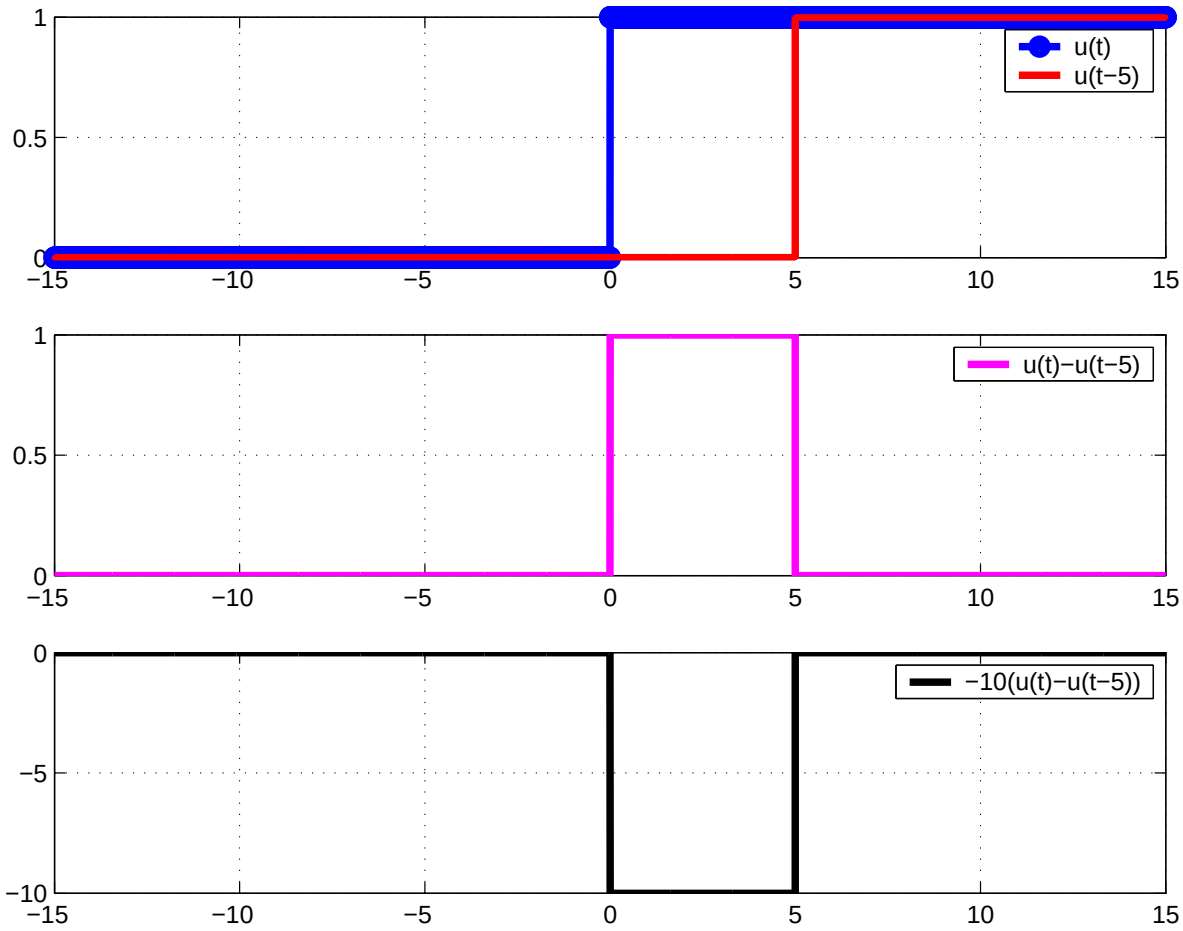
$$f\left(\frac{-3t+1}{5}\right) = f\left(\frac{3}{5}(-t+\frac{1}{3})\right)$$



Exercise 4 (1)

Express $f(t)$ as a combination of amplitude scaled and time shifted versions of $u(t)$:

$$f(t) = -10\,[u(t) - u(t-5)]$$



Exercise 4 (2-a)

Express $f(t)$ as an amplitude scaled, time scaled and time shifted version of $\mathrm{rect}(t)$.

Let:

$$f(t) = a\mathrm{rect}(b(t+c))$$

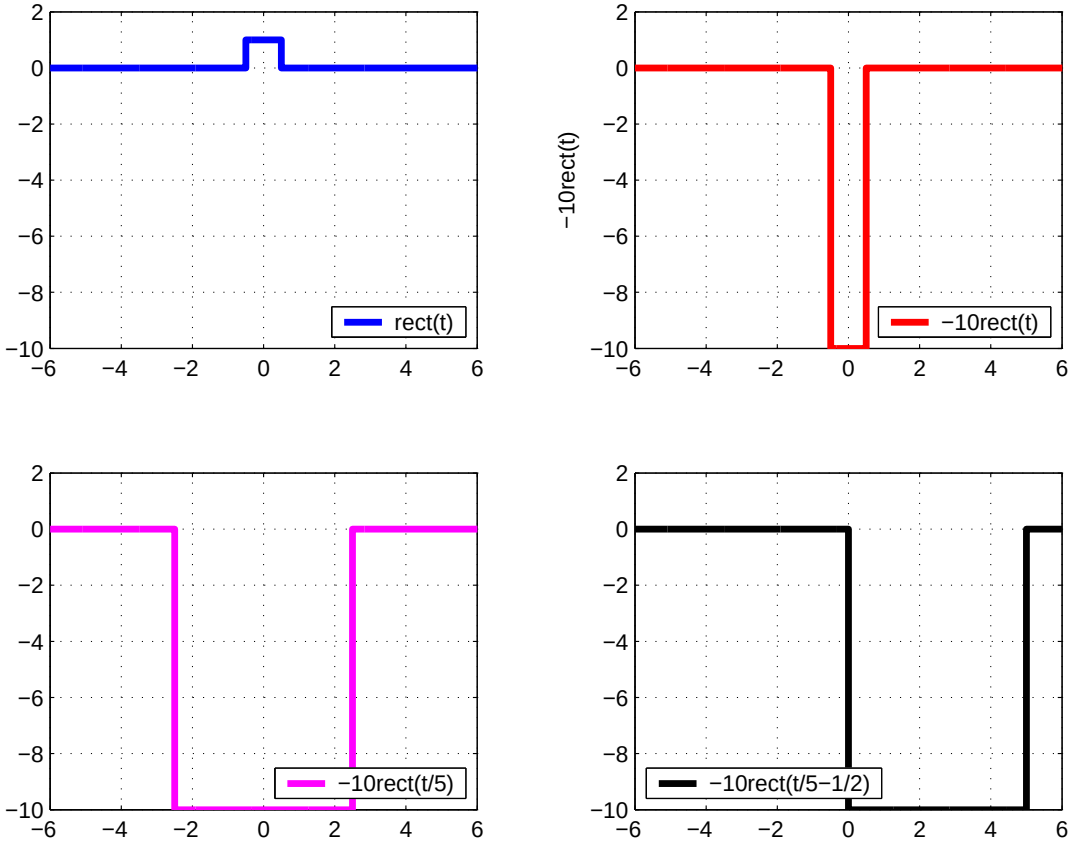
and determine the values of a, b and c.

- Given that the non-zero amplitude of $f(t)$ is equal to -10, then $a = -10$.
- Given that the duration of $f(t)$ is equal to 5 (that is 5 times the duration of the unit rectangular function), then $b = \frac{1}{5}$.
- Given that the center of $f(t)$ is at $t = \frac{5}{2}$, then $c = -\frac{5}{2}$.

Consequently:

$$f(t) = -10\mathrm{rect}\left(\frac{1}{5}(t - \frac{5}{2})\right) = -10\mathrm{rect}\left(\frac{t}{5} - \frac{1}{2}\right)$$

Exercise 4 (2-b)



Exercise 5 (1)

- Consider the function $u(at)$ where $a > 0$:

$$u(t) = \begin{cases} 1, & t \geq 0; \\ 0, & t < 0. \end{cases}$$

$$\Rightarrow \; u(at) = \begin{cases} 1, & at \geq 0; \\ 0, & at < 0. \end{cases} = \begin{cases} 1, & t \geq 0; \\ 0, & t < 0. \end{cases}$$

  implying that $u(at) = u(t)$ for $a > 0$.
- In other words, $u(t)$ is invariant with time scaling (if the scaling factor is positive).

Exercise 5 (2)

- Consider the function $\mathrm{ramp}(at)$ where $a > 0$:

$$\mathrm{ramp}(t) = \begin{cases} t, & t \geq 0; \\ 0, & t < 0. \end{cases}$$

$$\Rightarrow \mathrm{ramp}(at) = \begin{cases} at, & at \geq 0; \\ 0, & at < 0. \end{cases} = \begin{cases} at, & t \geq 0; \\ 0, & t < 0. \end{cases}$$

$$= a \begin{cases} t, & t \geq 0; \\ 0, & t < 0. \end{cases}$$

implying that $\mathrm{ramp}(at) = a\mathrm{ramp}(t)$ for $a > 0$.

Exercise 6 (1)

- The function $\mathrm{rect}(t)$ is centered at 0 and has a duration of 1.
- Since $f(t)$ is centered at 4 and has a duration of 2:

$$f(t) = \mathrm{rect}\left(\frac{t-4}{2}\right)$$

and in terms of the unit step function:

$$f(t) = u(t-3) - u(t-5)$$

- Note that since $\mathrm{rect}(t) = u(t+\frac{1}{2}) - u(t-\frac{1}{2})$:

$$\begin{aligned} f(t) = \mathrm{rect}\left(\frac{t-4}{2}\right) &= u\left(\frac{t-4}{2}+\frac{1}{2}\right) - u\left(\frac{t-4}{2}-\frac{1}{2}\right) \\ &= u\left(\frac{t-3}{2}\right) - u\left(\frac{t-5}{2}\right) \\ &= u(t-3) - u(t-5) \end{aligned}$$

where the last equality follows since $u(t)$ is invariant with time scaling (refer to problem 5).

Exercise 6 (2)

- On the other hand, $g(t)$ is centered at $9/4$ and has a duration of $1/2$:

$$g(t) = 2\mathrm{rect}\left(\frac{t-9/4}{1/2}\right) = 2\mathrm{rect}\left(2\left(t-\frac{9}{4}\right)\right) = 2\mathrm{rect}\left(2t-\frac{9}{2}\right)$$

  and in terms of the unit step function:

$$g(t) = 2\left[u(t-2) - u(t-5/2)\right]$$

- Note that since $\mathrm{rect}(t) = u(t+\frac{1}{2}) - u(t-\frac{1}{2})$:

$$\begin{aligned} g(t) &= 2\mathrm{rect}\left(2t-\frac{9}{2}\right) = 2u\left(2t-\frac{9}{2}+\frac{1}{2}\right) - 2u\left(2t-\frac{9}{2}-\frac{1}{2}\right) \\ &= 2u\left(2t-4\right) - u\left(2t-5\right) \\ &= 2u(2(t-2)) - 2u(2(t-5/2)) = 2u(t-2) - 2u(t-5/2) \end{aligned}$$

  where the last equality follows since $u(t)$ is invariant with time scaling (refer to problem 5).

Exercise 7 (1)

- The function $\mathrm{tri}(t)$ is centered at 0 and has a duration of 2.
- Since $f(t)$ is centered at 3 and has a duration of 1:

$$f(t) = 2\mathrm{tri}\left(2(t-3)\right)$$

- The transitions in $f(t)$ are occurring at $5/2$, 3 and $7/2$:
  - At $t = 5/2$: the slope passes from 0 to $\frac{2}{1/2} = 4$.
    - The change in slope is 4.
  - At $t = 3$: the slope passes from $+4$ to -4.
    - The change in slope is -8.
  - At $t = 7/2$: the slope passes from -4 to 0.
    - The change in slope is 4.
- Consequently:

$$f(t) = 4\mathrm{ramp}\left(t - \frac{5}{2}\right) - 8\mathrm{ramp}\left(t-3\right) + 4\mathrm{ramp}\left(t - \frac{7}{2}\right)$$

Exercise 7 (2)

- Note that since $\mathrm{tri}(t) = \mathrm{ramp}(t+1) - 2\mathrm{ramp}(t) + \mathrm{ramp}(t-1)$ (from the lecture):

$$
\begin{aligned}
f(t) &= 2\mathrm{tri}\left(2(t-3)\right) \\
&= 2\mathrm{ramp}(2(t-3)+1) - 4\mathrm{ramp}(2(t-3)) + 2\mathrm{ramp}(2(t-3)-1) \\
&= 2\mathrm{ramp}(2t-5) - 4\mathrm{ramp}(2(t-3)) + 2\mathrm{ramp}(2t-7) \\
&= 2\mathrm{ramp}\left(2(t-5/2)\right) - 4\mathrm{ramp}(2(t-3)) + 2\mathrm{ramp}\left(2(t-7/2)\right) \\
&= 4\mathrm{ramp}\left(t-\frac{5}{2}\right) - 8\mathrm{ramp}\left(t-3\right) + 4\mathrm{ramp}\left(t-\frac{7}{2}\right)
\end{aligned}
$$

since $\mathrm{ramp}(at) = a\mathrm{ramp}(t)$ for $a > 0$ (refer to exercise 5).

Exercise 8

- The transitions in $f(t)$ are occurring at 0, 1, 3, 4 and 5:
  - At $t = 0$, $f(t)$ is undergoing a variation of $3 - 0 = 3$.
  - At $t = 1$, $f(t)$ is undergoing a variation of $-2 - 3 = -5$.
  - At $t = 3$, $f(t)$ is undergoing a variation of $0 - (-2) = 2$.
  - At $t = 4$, $f(t)$ is undergoing a variation of $1 - 0 = 1$.
  - At $t = 5$, $f(t)$ is undergoing a variation of $0 - 1 = -1$.
- Consequently:

$$f(t) = 3u(t) - 5u(t-1) + 2u(t-3) + u(t-4) - u(t-5)$$

- For example, at $t = 2$:

$$\begin{aligned} f(2) &= 3\underbrace{u(2)}_{=1} - 5\underbrace{u(1)}_{=1} + 2\underbrace{u(-1)}_{=0} + \underbrace{u(-2)}_{=0} - \underbrace{u(-3)}_{=0} \\ &= 3 - 5 = -2 \ \checkmark \end{aligned}$$

Exercise 9 (a)

- The energy of $\mathrm{tri}(t)$ is given by:

$$
\begin{aligned}
E_1 &= \int_{-\infty}^{+\infty} \mathrm{tri}^2(t)\mathrm{d}t && (1)\\
&= \int_{-1}^{+1} \mathrm{tri}^2(t)\mathrm{d}t && (2)\\
&= 2\int_{0}^{+1} \mathrm{tri}^2(t)\mathrm{d}t && (3)\\
&= 2\int_{0}^{+1} (1-t)^2\mathrm{d}t && (4)\\
&= 2\left[-\frac{1}{3}(1-t)^3\right]_0^1 = \frac{2}{3} \quad (\mathrm{J})
\end{aligned}
$$

  - Eq. (2) follows since $\mathrm{tri}(t) = 0$ for $|t| > 1$.
  - Eq. (3) follows since $\mathrm{tri}^2(t)$ is even.
  - Eq. (4) follows since $\mathrm{tri}(t) = 1 - t$ for $0 \leq t \leq 1$.

Exercise 9 (b)

- The energy of $\mathrm{tri}(t) + 5\mathrm{tri}(t-4)$ is given by:

$$E_2 = \int_{-\infty}^{+\infty} [\mathrm{tri}(t) + 5\mathrm{tri}(t-4)]^2 \, \mathrm{d}t$$

$$= \underbrace{\int_{-\infty}^{+\infty} [\mathrm{tri}(t)]^2 \, \mathrm{d}t}_{I_1} + 25 \underbrace{\int_{-\infty}^{+\infty} [\mathrm{tri}(t-4)]^2 \, \mathrm{d}t}_{I_2} + 10 \underbrace{\int_{-\infty}^{+\infty} \mathrm{tri}(t)\mathrm{tri}(t-4) \mathrm{d}t}_{I_3}$$

  - $I_1 = E_1 = 2/3$ from part (a).
  - $I_2 = E_1 = 2/3$. In fact, performing the change of variable $t' = t - 4$ results in $I_2 = \int_{-\infty}^{+\infty} [\mathrm{tri}(t')]^2 \, \mathrm{d}t' = E_1$.
    - Note that time-shifting does not change the energy of a signal; i.e., $\mathrm{tri}(t)$ and $\mathrm{tri}(t-4)$ have the same energy E_1.
  - $I_3 = 0$ since $\mathrm{tri}(t)\mathrm{tri}(t-4) = 0$. In fact, there is no overlap between $\mathrm{tri}(t)$ (that is different from zero for $-1 < t < 1$) and $\mathrm{tri}(t-4)$ (that is different from zero for $3 < t < 5$).
- Finally:

$$E_2 = 26E_1 = 17.33 \quad (\mathrm{J})$$